

Debao Huang

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Profile: Ph.D. candidate specializing in large-scale 3D/4D reconstruction, with experience in real-world deployments

Research Interests: 3D Computer Vision, Photogrammetry, Machine Learning, Deep Learning

Education

Ph.D., Electrical and Computer Engineering, The Ohio State University, USA (Advisor: Prof. Rongjun Qin) 08/2026 (expected)
M.S., Electrical and Computer Engineering, The Ohio State University, USA 05/2023
B.S., Telecommunication Engineering, Sun Yat-sen University, China 06/2019

Technical Skills

Programming Languages: C++, Python, MATLAB

Skills: Large-scale 3D Reconstruction (SfM, NeRF, 3DGS, 4DGS), Vision Foundation Models, Generative AI, Multimodal Learning

Frameworks and Libraries: PyTorch, Diffusers, Nerfstudio, PEFT, Docker, OpenCV, Ceres, COLMAP

Certifications: FAA Remote Pilot. Executed survey missions totaling 73.4 hectares and 47.8 km of flight path.

Research Experience

Intrinsic Image Decomposition for Realism Enhancement of Digital Twins and 3D Assets – ONR 01/2025–Present

- Constructed a 33K+ image real-world dataset with pixel-aligned albedo, shading, normal, and metric depth using a physics-based inverse-rendering pipeline, enabling supervised intrinsic image decomposition.
- Fine-tuned Stable Diffusion models using LoRA to close sim-to-real gap, improving PSNR by 7 dB over SOTA and enabling downstream applications such as relighting and material editing. (outcomes: **P1: CVPR 2026**).

Live 3D Gaussian Splatting for Intelligence, Surveillance, and Reconnaissance – AFRL 10/2024–10/2025

- Developed a novel 4DGS pipeline to reconstruct dynamic urban scenes from monocular UAV video, achieving an improvement of 4 dB in PSNR over SOTA for rendering moving objects.
- Integrated 4DGS with photogrammetry, video segmentation and tracking, monocular depth estimation, and physics-guided trajectory optimization into a unified reconstruction framework. (outcomes: **P3**).

Enabling Seamless 3D Semantic Reconstruction from Heterogeneous Data at Scale – ONR 01/2021–02/2025

- Developed SfM algorithms and introduced novel geometric constraints in bundle adjustment for multi-camera systems, enhancing the accuracy of camera pose estimation by up to 86%. (outcomes: **S1, H1, P4**).
- Developed a scalable Multi-Camera Tiling (MCT) NeRF framework for large-scale aerial datasets, reducing VRAM usage by 15.2% and significantly improving geometric completeness for fine structures. (outcomes: **P5**).

Software & System Development

[S1] Software: MetricSfM (Lead Developer) | C++, OpenMP, CUDA, Ceres, Eigen 2024

- Engineered an end-to-end Structure-from-Motion pipeline supporting heterogeneous data sources (aerial, UAV, GoPro, mobile, and underwater cameras) with both GUI and CLI interfaces. **Deployed to the Office of Naval Research.**

[H1] Hardware: Multi-camera Mobile Mapping Systems 2022

- Developed a low-cost ground-mapping system using arbitrarily positioned GoPro cameras, incorporating geometric constraints for multi-camera self-calibration in MetricSfM to achieve robust 3D reconstruction.

Selected Publications

[P1] Song, S.[†], **Huang, D.**[†], Deng, D., Xiong, H., Tang, Y., Zhao, Y., & Qin, R. (2026). "Olbedo: An Albedo and Shading Aerial Dataset for Large-Scale Outdoor Environments". *The IEEE/CVF Conference on Computer Vision and Pattern Recognition*. (**CVPR 2026**).

[P2] **Huang, D.**, Qin, R. (2026). "Uncertainty Quantification Framework for Aerial and UAV Photogrammetry through Error Propagation". *ISPRS Journal of Photogrammetry and Remote Sensing*. (**Top-tier journal, IF:12.2**).

[P3] **Huang, D.**[†], Liu, H.[†], Xu, N., & Qin, R. (2025). "Dynamic Urban Scene Modeling with 3D Gaussian Splatting from UAV Full Motion Videos". *ISPRS Geospatial Week*.

[P4] **Huang, D.**, Qin, R., & Elhashash, M. (2024). "Bundle Adjustment with Motion Constraints for Uncalibrated Multi-camera Systems at The Ground Level". *ISPRS Journal of Photogrammetry and Remote Sensing*. (**Top-tier journal, IF:12.2**).

[P5] Xu, N., Qin, R., **Huang, D.**, & Remondino, F. (2024). "Multi-tiling Neural Radiance Field (NeRF)—Geometric Assessment on Large-scale Aerial Datasets". *The Photogrammetric Record*. (**Cover article of 12/2024 issue**).

Honors & Media Coverage

- Recipient of **Lyman J. Ladner Memorial Scholarship**, American Society for Photogrammetry and Remote Sensing, 2026. Awarded to outstanding students (**Top 1%**) pursuing research and careers in photogrammetry.
- Research on 3D reconstruction using satellite data for disaster response, featured in **Ohio State News** [Link].